

# Wakacyjne Wyzwanie

Z  KN Solvro

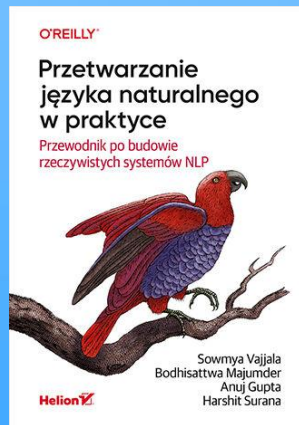
# NLP – Przetwarzanie Języka Naturalnego



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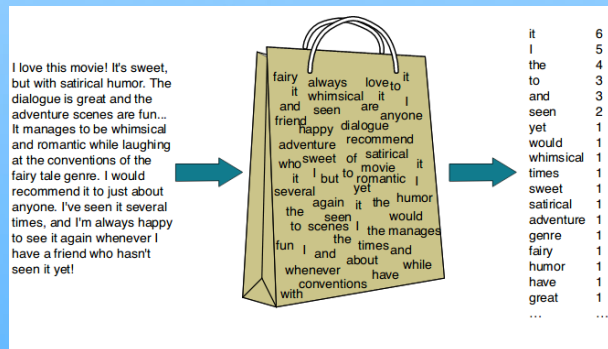
# Czym jest enelpi ?

- **Przetwarzanie języka naturalnego** (ang. natural language processing, NLP) – interdyscyplinarna dziedzina, łącząca zagadnienia sztucznej inteligencji i językoznawstwa, zajmująca się automatyzacją analizy, rozumienia, tłumaczenia i generowania języka naturalnego przez komputer



# Reprezentacje tekstu kiedyś i dzisiaj

## Bag of words

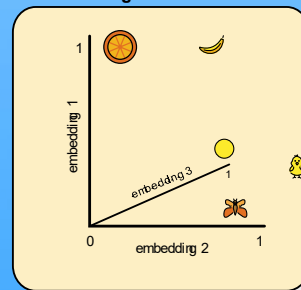


## Osadzenia - Embeddingi

### Vocab

1. banana
2. orange
3. yellow
4. finch
5. monarch

### Embedding



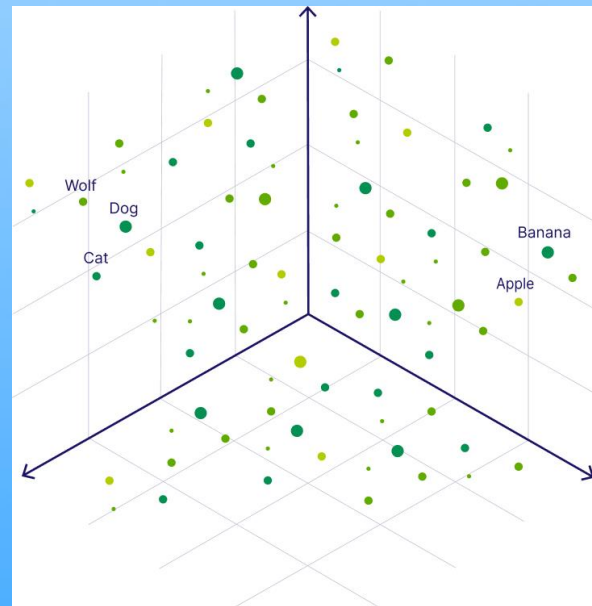
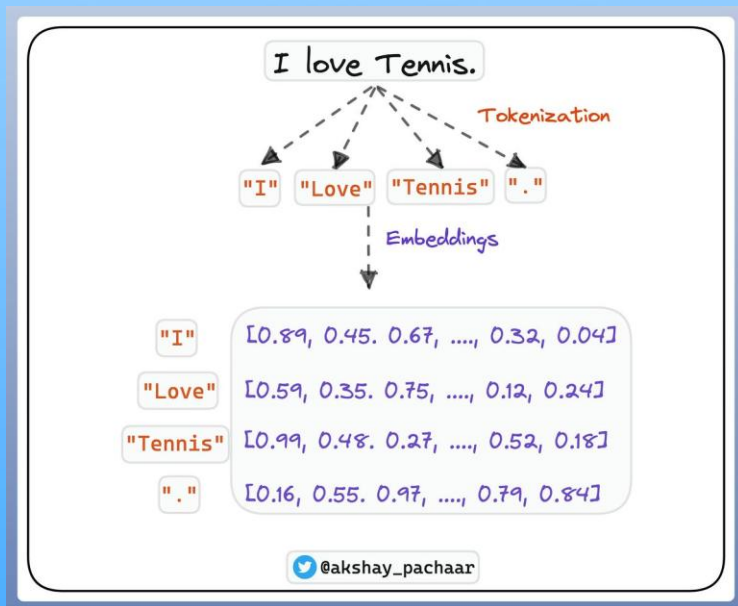
| $e_1$ | $e_2$ | $e_3$ |
|-------|-------|-------|
| 0.91  | 0.13  | 0.89  |
| 0.98  | 0.08  | 0.03  |
| 0.11  | 0.01  | 0.99  |
| 0.15  | 0.98  | 0.92  |
| 0.12  | 0.91  | 0.09  |

Embedding 1 could be fruit  
Embedding 2 could be animal  
Embedding 3 could be color



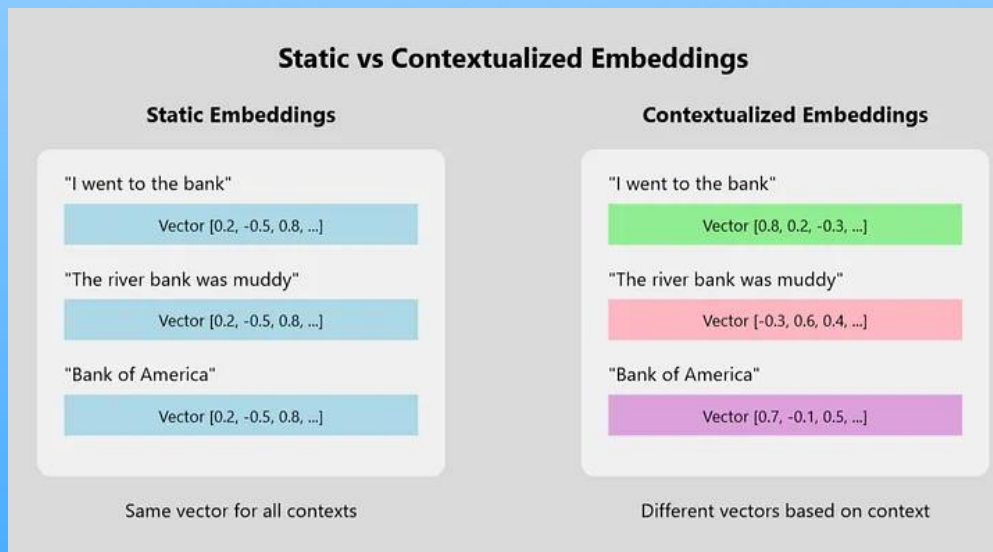
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# Embeddingi i tokenizacija



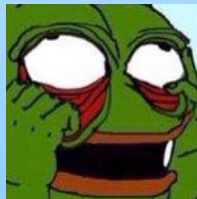
# Embeddingi statyczne vs kontekstowe

Problem, który został rozwiązany przez wprowadzenie kontekstu



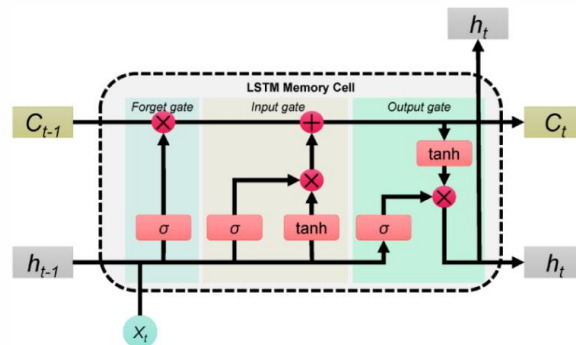
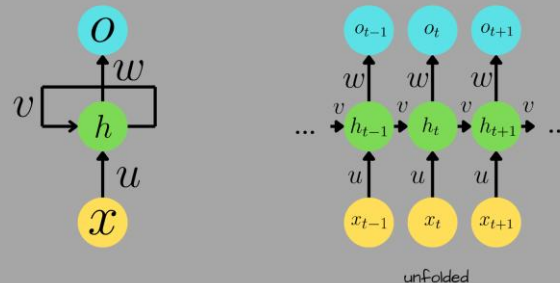


# Krótko o LSTM



- **Problem:** tekst to sekwencja - potrzebujemy modelu, który "pamięta" co było wcześniej
- **Intuicja RNN:** sieć czyta słowo po słowie, przekazując "stan ukryty" (pamięć) do następnego kroku - jak czytanie zdania i trzymanie w głowie kontekstu
- **Problem RNN:** zanikający gradient, słaba pamięć długodystansowa - stąd LSTM
- **Kluczowe do zapamiętania:** RNN/LSTM czyta sekwencyjnie, jednokierunkowo (lub dwa oddzielne przebiegi) - to będzie ważne za chwilę

*RNN - architecture*



# Ulica Sezamkowa



Ulica Sezamkowa

1969 · Musical · 56 sezonów



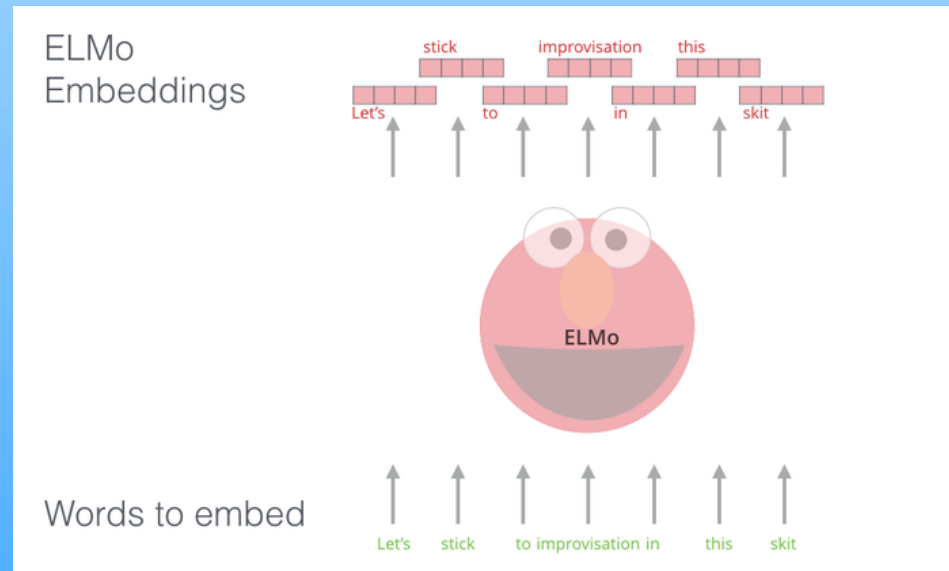
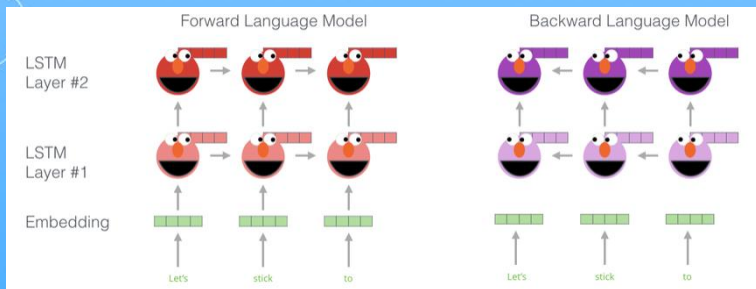
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# ELMo - Embeddings from Language Models



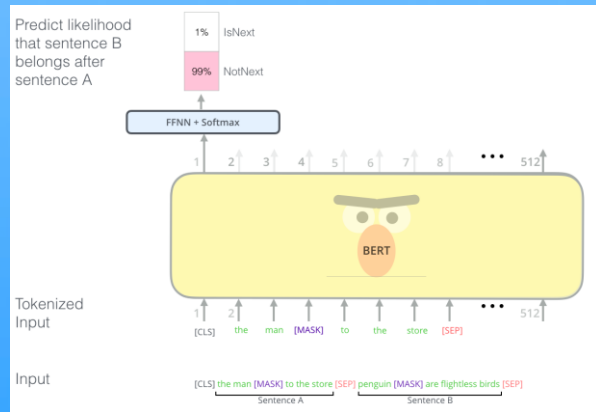
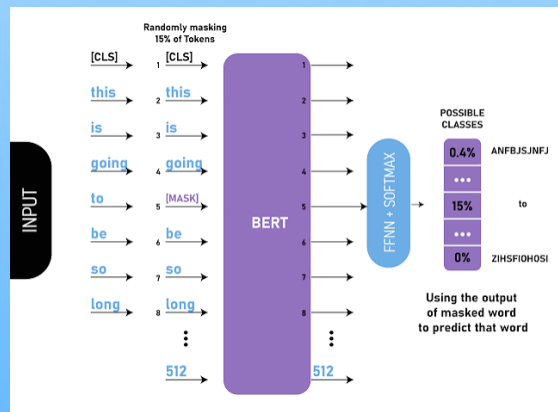
- **ELMo** pozwala na wydobywanie embeddingów kontekstu co diametralnie zmieniło zrozumienie języka przez modele i wykorzystanie zależności tekstu.



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# BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

- **Kluczowa różnica względem ELMo:** prawdziwa dwukierunkowość dzięki transformerowi
- **Masked Language Modeling (MLM):** losowo maskujemy słowa w zdaniu i model uczy się je przewidywać na podstawie całego kontekstu
- **Next Sentence Prediction** - jako drugi cel treningowy
- **Do czego się nadaje ?** Lasyfikacja tekstu, embeddingi zdań/dokumentów, wyszukiwanie semantyczne (embedding + podobieństwo wektorowe) i tak dalej

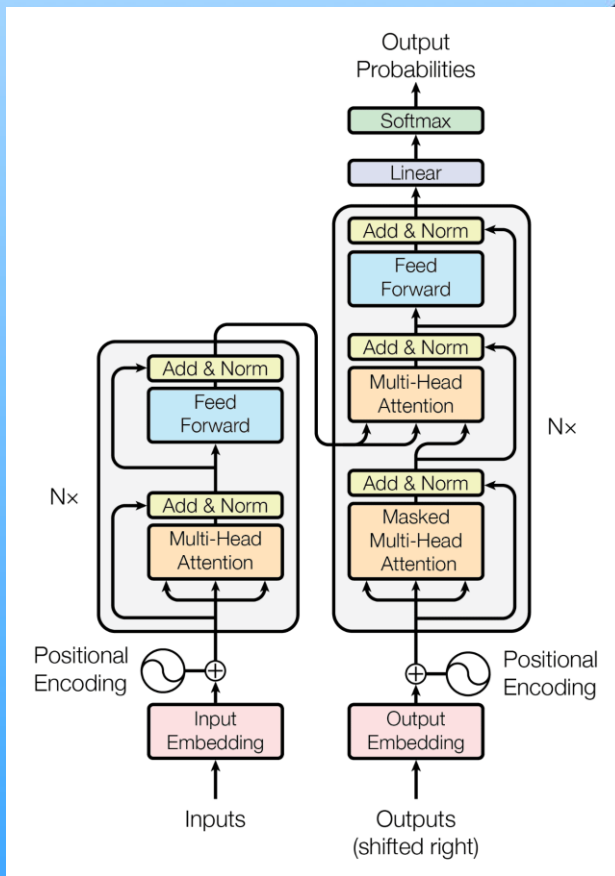


# Attention is all you need (serio)

- Model zwany Transformerem zaprezentowany w paperze **Attention Is All You Need** zrewolucjonizował dziedzinę sztucznej inteligencji i dał jej potężnego kopa – to właśnie ten moduł dał życie dużym modelom językowym – **LLM**om. **Wszystko przez mechanizm nazwany uwagą (attention)**



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# Jak działa mechanizm uwagi ?

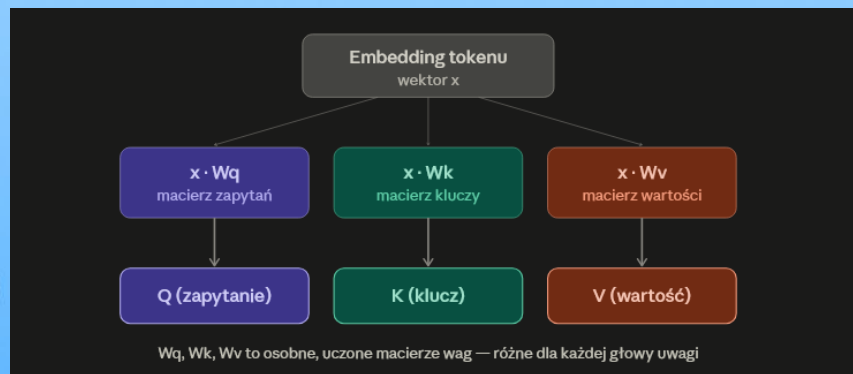


|         | Dwóch | polaków | w   | teamie | Wander | Amazing |
|---------|-------|---------|-----|--------|--------|---------|
| Dwóch   | .30   | .25     | .10 | .15    | .10    | .10     |
| polaków | .10   | .35     | .05 | .10    | .15    | .25     |
| w       | .15   | .10     | .40 | .15    | .10    | .10     |
| teamie  | .10   | .15     | .10 | .35    | .15    | .15     |
| Wander  | .05   | .35     | .03 | .07    | .15    | .35     |
| Amazing | .05   | .30     | .05 | .10    | .20    | .30     |

Każdy wiersz to token zadający pytanie (query), każda kolumna to token dający odpowiedź (key), a liczba w komórce to waga uwagi (im ciemniejszy/bardziej intensywny odcień, tym silniejsze powiązanie)



# Trochę bardziej formalnie



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

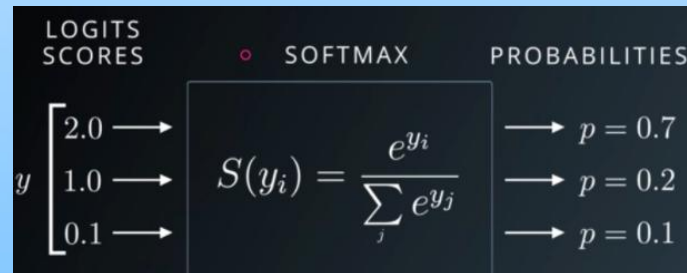


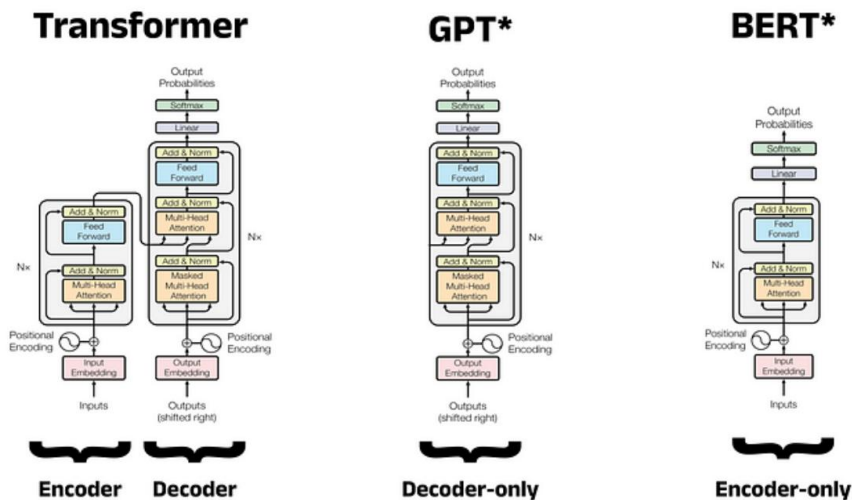
Diagram illustrating the matrix operations:

- Input matrices:  $Q$  (purple, 3x3) and  $K^T$  (orange, 3x3)
- Operation:  $Q \times K^T$
- Result:  $\frac{Q \times K^T}{\sqrt{d_k}}$
- Operation:  $\text{softmax}(\cdot)$
- Result:  $Z$  (pink, 3x3)
- Operation:  $Z \times V$  (blue, 3x3)
- Final result:  $V$  (blue, 3x3)

# Enkoder vs Dekoder vs Enkoder-Dekoder



| Feature                  | GPT                            | BERT                            | T5                                |
|--------------------------|--------------------------------|---------------------------------|-----------------------------------|
| <b>Main Use</b>          | Text generation                | Text understanding              | Multi-purpose NLP tasks           |
| <b>How it reads text</b> | Left to right (unidirectional) | Both directions (bidirectional) | Both + decoder                    |
| <b>Best for...</b>       | Chatbots, content creation     | Search engines, classification  | Summarization, translation, Q&A   |
| <b>Biggest Weakness</b>  | Can be inaccurate at times     | Can't generate text             | Requires a lot of computing power |





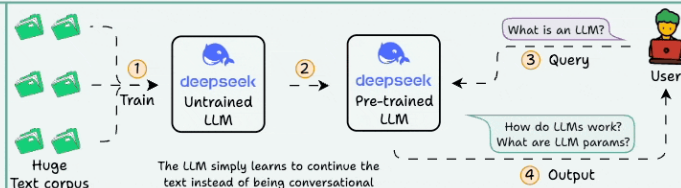
# 4 stages of LLM Training

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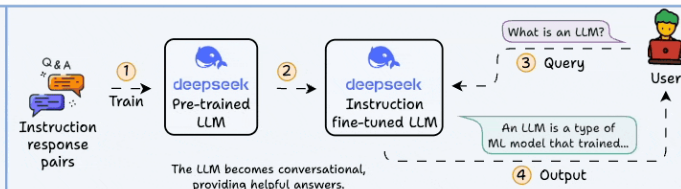
## Stage 0 Randomly initialized model



## Stage 1 Pre-training



## Stage 2 Instruction fine-tuning



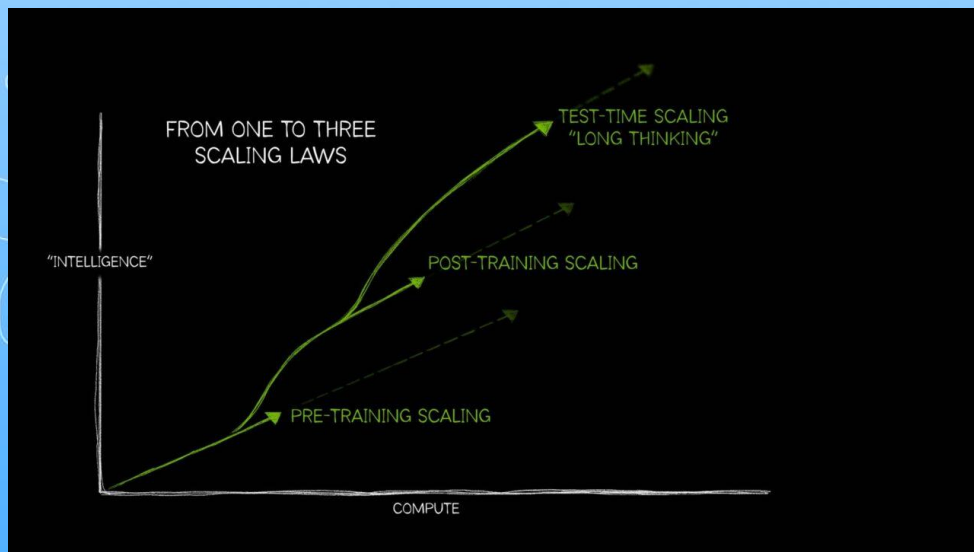
## Stage 3 Preference Fine-tuning



## Stage 4 Reasoning fine-tuning



# Prawo Moore'a dla LLMów



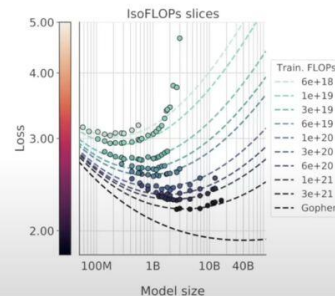
## LLM Scaling Laws

Performance of LLMs is a smooth, well-behaved, predictable function of:

- $N$ , the number of parameters in the network
- $D$ , the amount of text we train on

And the trends do not show signs of "topping out"

=> We can expect more intelligence "for free" by scaling



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# Agent Arena

🏆 Overall

Dynamic ranking of models on how well they orchestrate tools for real-world agentic tasks, based on signals like tool reliability, task completion, and steerability.

🕒 Aug 19, 2026    ✅ 1,878,054 sessions    📋 49 models



🔍 Hide Filters

Cost

Output Tokens

50th percentile



## Pareto Frontier

Net Improvement x Cost/task from real Agent Mode tasks

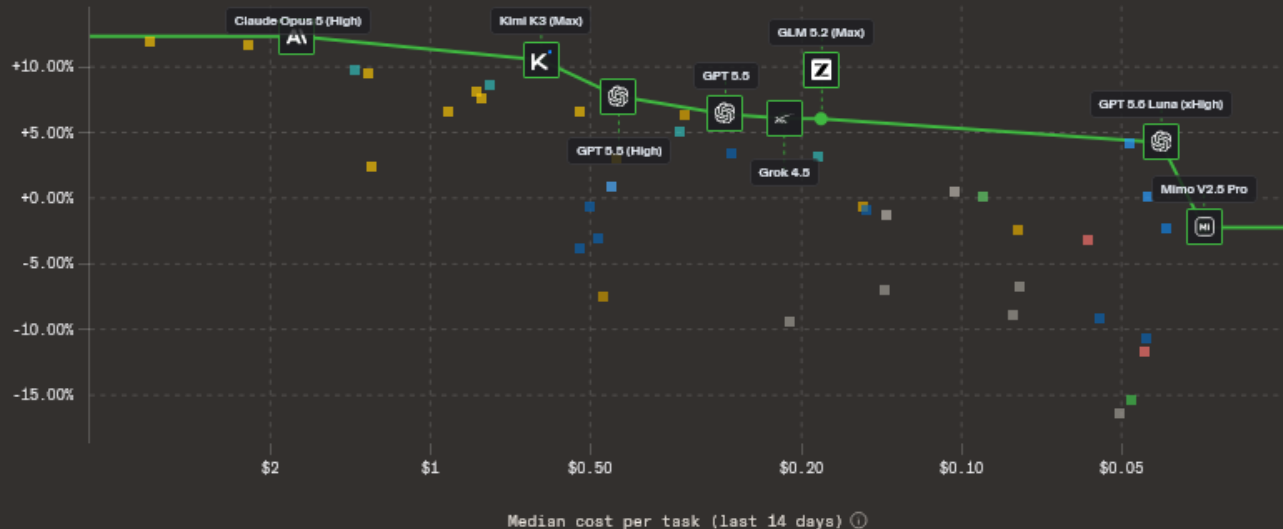
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## Pareto Optimal Models

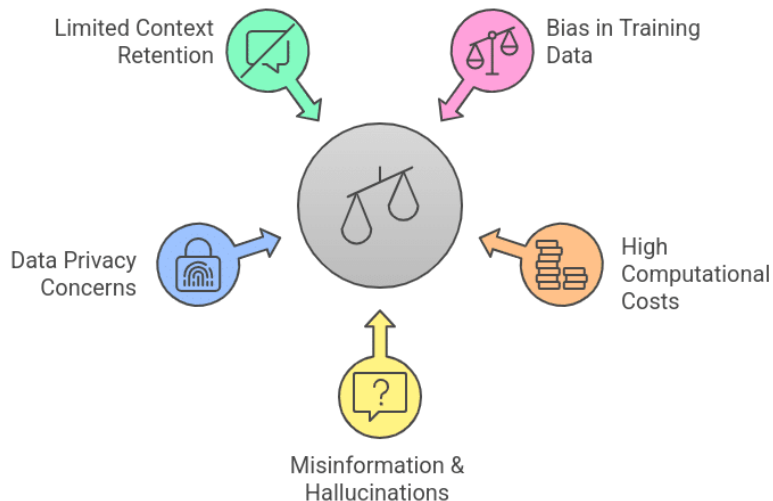
Net Improvement x Cost/task



|    |                            |             |
|----|----------------------------|-------------|
| AI | Claude Opus 5 (High)       | +12.34%     |
|    | Anthropic - Proprietary    | \$1.78/task |
| K  | Kimi K3 (Max)              | +10.53%     |
|    | Moonshot - Kimi K3 lice... | \$0.62/task |
|    | GPT 5.5 (High)             | +7.75%      |
|    | OpenAI - Proprietary       | \$0.44/task |
|    | GPT 5.5                    | +6.40%      |
|    | OpenAI - Proprietary       | \$0.28/task |
|    | Grok 4.5                   | +6.08%      |
|    | SpaceXAI - Proprietary     | \$0.22/task |
| Z  | GLM 5.2 (Max)              | +6.06%      |
|    | Z.ai - MIT                 | \$0.18/task |
|    | GPT 5.6 Luna (xHigh)       | +4.25%      |
|    | OpenAI - Proprietary       | \$0.04/task |
| MI | Mimo V2.5 Pro              | -2.22%      |
|    | Xiaomi - MIT               | \$0.04/task |

# Problemy, problemiki LLMów

## Challenges of Large Language Models



What happened on June 4, 1989 at Tiananmen Square?



Sorry, that's beyond my current scope. Let's talk about something else.



What's the best espresso machine under \$200 that comes close to the taste of coffee in Italy?

If you're aiming to recreate the ambience of an Italian café at home without exceeding a \$200 budget, there are several espresso machines that can deliver rich, authentic flavors. Below are some top picks that balance quality and affordability:



**Versatile Choice**  
De'Longhi ECP3220 Espresso Machine  
\$178.95  
De'Longhi and 8 others



**Compact Design**  
De'Longhi Compact Professional Espresso Machine  
\$124.99  
De'Longhi and 3 others



**Best Budget Choice**  
De'Longhi Stilosa Manual Espresso Machine  
\$99.95  
Purilio



**Best Budget Choice**

**De'Longhi Stilosa Manual Espresso Machine**

★★★★★ 4.3 (412)

### Purchasing options

**De'Longhi** \$99.95 [Buy](#)  
De'Longhi Stilosa Manual Espresso Machine  
Free delivery by Wed, Apr 30 - 30-day return

**Walmart** \$99.99 [Buy](#)  
De'Longhi Stilosa Manual Espresso Machine  
Free delivery

**BrandsMart USA** \$89.99 [Buy](#)  
De'Longhi Stilosa Manual Espresso Machine  
Free delivery by Thu, Apr 29 - 90-day return



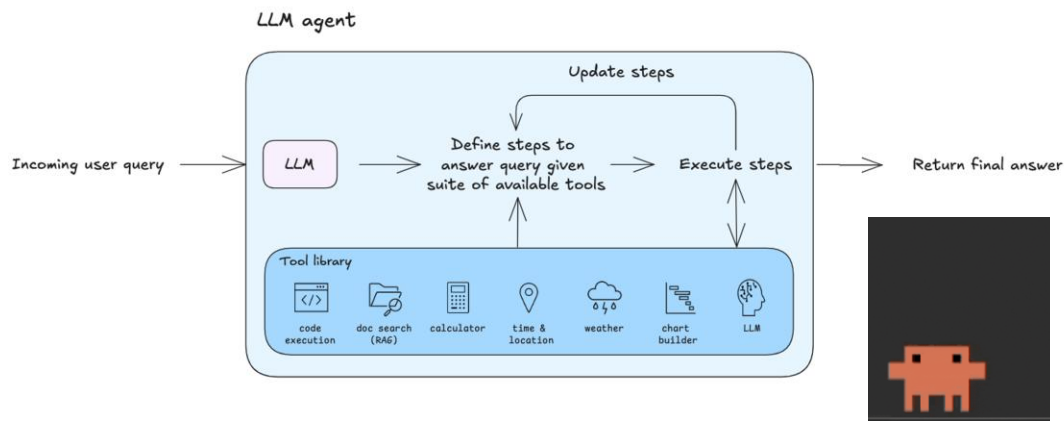
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# Agent – LLM jako system



Agent to inaczej LLM, który:

- Posiada zestaw narzędzi (toole)
- Działa w pętli wykonawczej
- Jest wyposażony w pamięć
- Wykonuje określone zadanie (lub zestaw zadań)

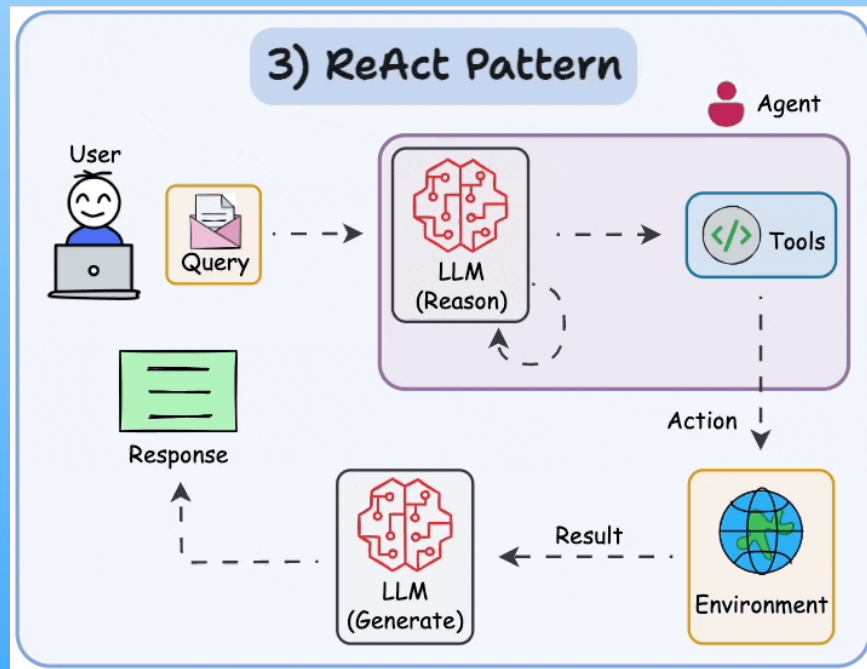


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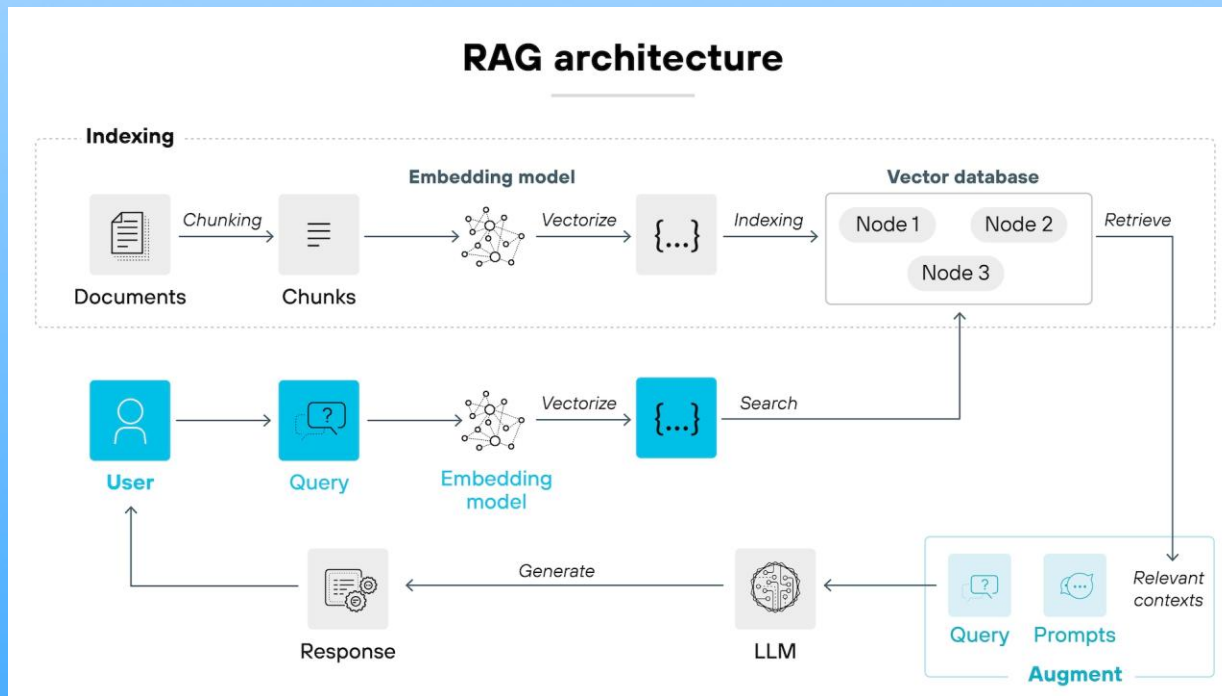
# ReAct – Reason and Act

Jest to połączenie techniki CoT (chain of thoughts - myślenia) oraz wykorzystania narzędzi. Wymyślone w 2022 roku – dalej jest podstawą systemów agentowych.





# RAG (Retrieval-Augmented Generation)



# LLM Wiki – A. Karpathy



## The Three-Layer Architecture

### Top Layer: The Schema (Rules)

[CLAUDE.md / AGENTS.md]

The configuration file. Dictates structure, conventions, and workflows. Co-evolved by human and LLM to enforce discipline.

Rules Flow Down

Data Flows Up

### Base Layer: Raw Sources (Immutable Truth)

Curated Source Material

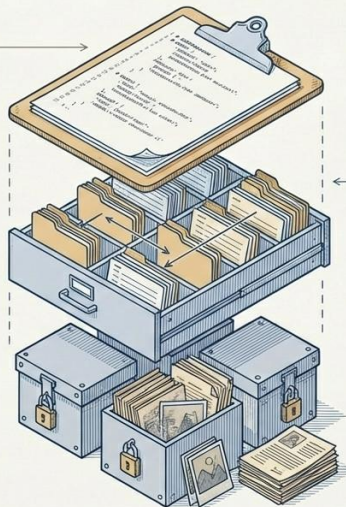
Your collection of PDFs, articles, and data. Read-only. The absolute source of truth.

### Middle Layer: The Wiki (Synthesis)

Directory of .md files

Summaries, entity pages, syntheses.

The LLM owns this layer entirely. You read it; the LLM writes it.



PRD.JD. 2024.RAG.WIKI.SPEC

NotebookLM



KN Solvra

Reinforcement  
Learning is terrible.

# OKF – Open Knowledge Format

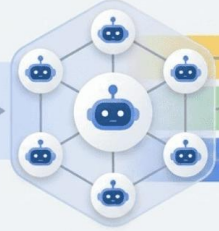


## The Open Knowledge Format (OKF): An Emerging Open Standard for AI Systems

### Standard Defined

```
OKF
version: 0.1
format: open-knowledge
...
title: Database Schema
type: schema
description: Schema for
customer data
tags: [bigquery, sales]
...
# Customer
```

### AI Processing Hub



### Structured Data



### Knowledge Graph



### Documents and Text-based Knowledge



Portable,  
Open/Flexible.



Simple,  
Human-readable.



Built for Humans  
and AI Systems.

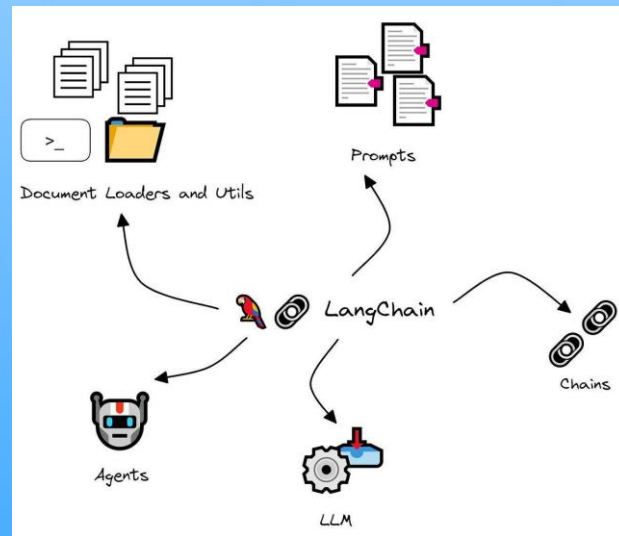
# Pythonowe frameworki do Agentic AI



## Quick-Reference Comparison: AI Agent Frameworks in 2026

Evaluated across 7 enterprise deployments in healthcare, logistics, fintech, and e-commerce (2025–2026)

| FRAMEWORK               | BEST FOR                                                      | PRODUCTION | LEARNING CURVE | HUMAN-IN-LOOP | COST CONTROL | BACKING                          |
|-------------------------|---------------------------------------------------------------|------------|----------------|---------------|--------------|----------------------------------|
| LangGraph               | Complex stateful workflows, compliance-heavy verticals        | ★★★★★      | Steep          | Native        | High         | LangChain (126K stars)           |
| CrewAI                  | Fast prototyping, role-based collaboration, content pipelines | ★★★★★      | Low            | Limited       | Medium       | CrewAI Inc.                      |
| AutoGen 2.0             | Research pipelines, Azure environments, async workflows       | ★★★★★      | Moderate       | Proxy pattern | Low          | Microsoft Research               |
| OpenAI Agents SDK       | OpenAI-native apps, rapid agent prototyping                   | ★★★★       | Low            | Basic         | Medium       | OpenAI                           |
| Anthropic Agent SDK     | Code agents, high-accuracy tasks, safety-critical workflows   | ★★★★★      | Moderate       | Built-in      | High         | Anthropic                        |
| Google ADK              | Vertex AI environments, multimodal agents                     | ★★★★       | Moderate       | Configurable  | Medium       | Google DeepMind                  |
| LlamaIndex Workflows    | RAG-heavy applications, knowledge retrieval agents            | ★★★★★      | Moderate       | Event-driven  | High         | LlamaIndex                       |
| Intuz Agentic Framework | Enterprise multi-agent orchestration, self-learning systems   | ★★★★★      | Managed        | Full          | High         | Intuz (16+ yrs, 1,700+ projects) |



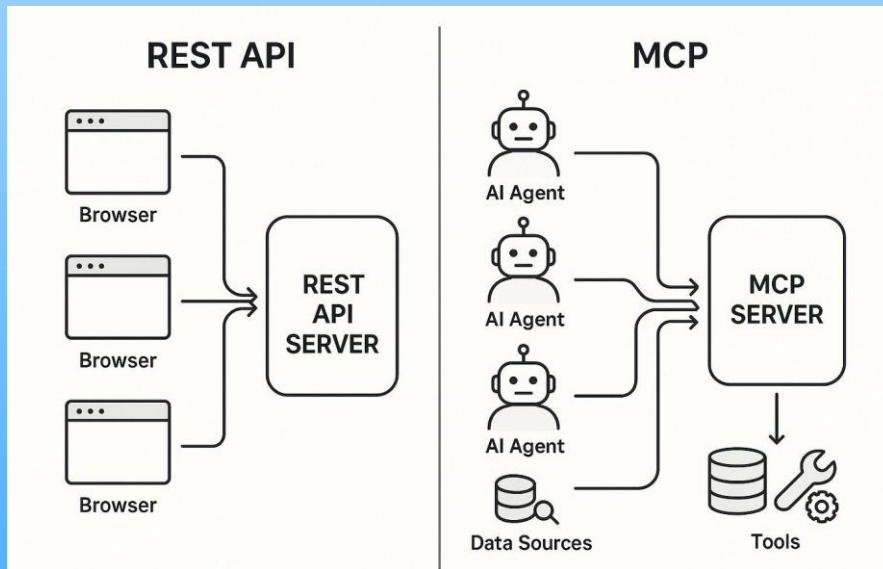
# MCP – Model Context Protocol

## Zalety MCP :

- Zarządzanie API
- Uniwersalny interfejs komunikacyjny między LLMami (takie USB-C)
- Łatwe wołanie funkcji i narzędzi
- Ogólnie przyjęty format

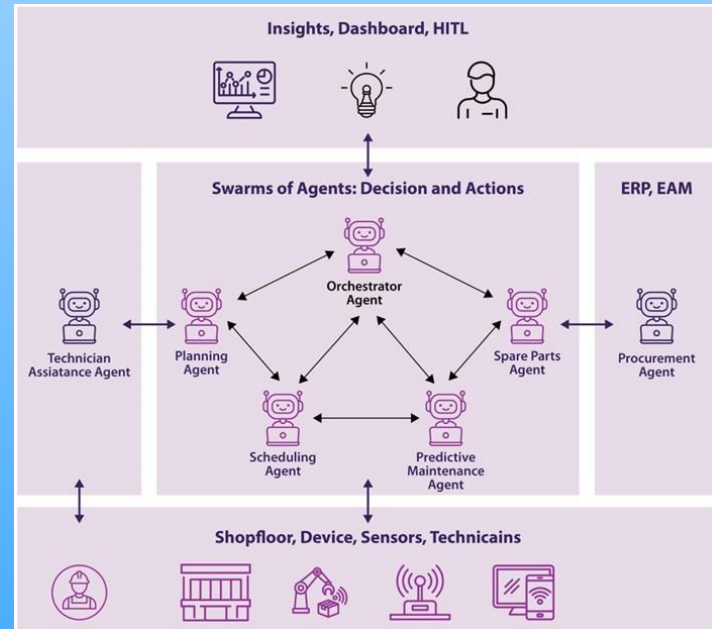
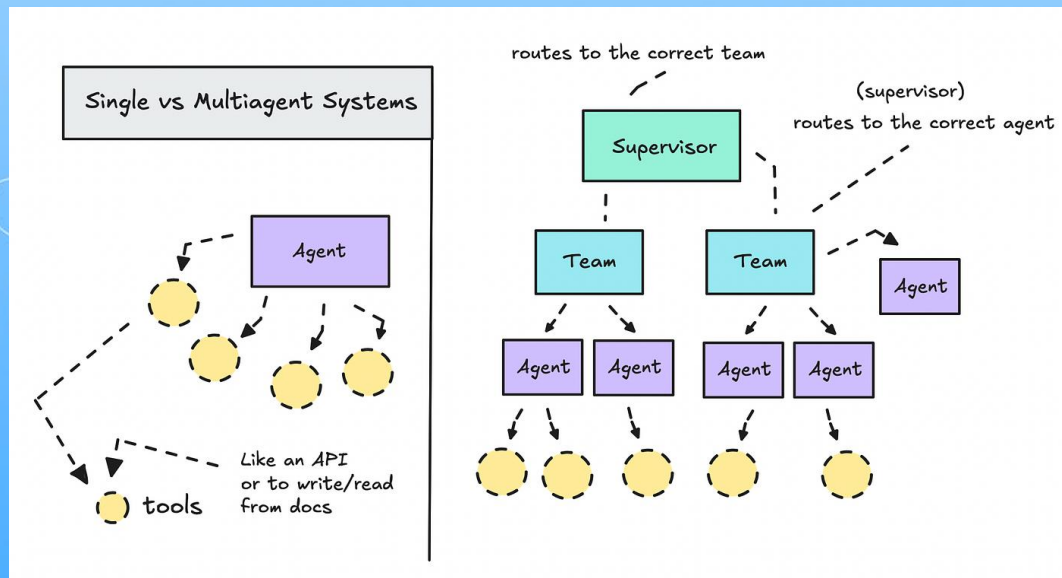


+ WandB  
+ HuggingFace





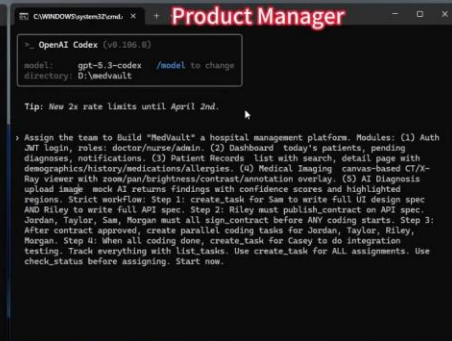
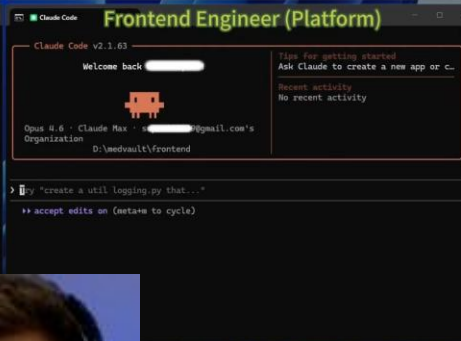
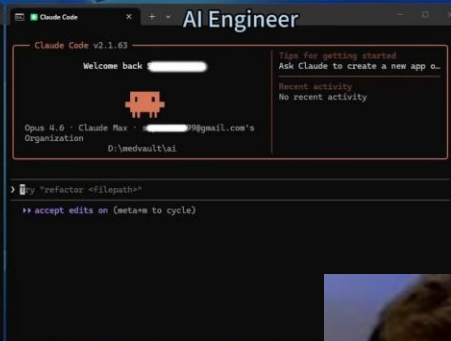
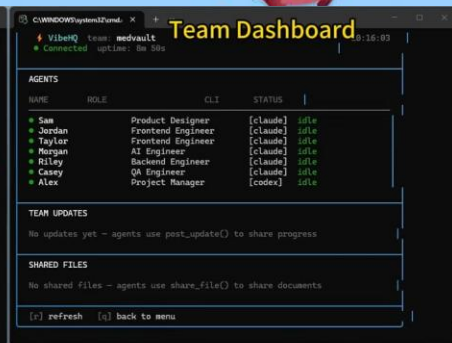
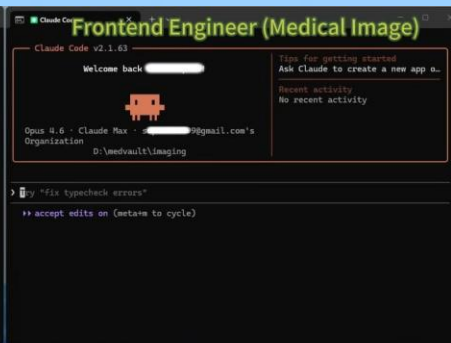
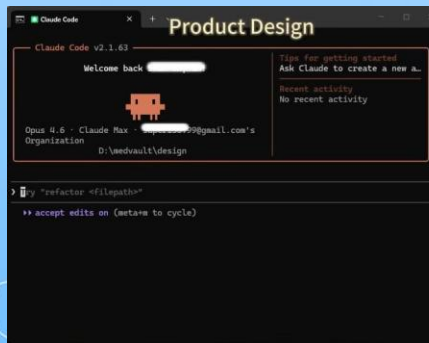
# Systemy wieloagentowe – agentowe korpo



POV: Your company's HR director is about to fire you on a Zoom call







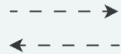
# A2A – Agent to Agent protocol



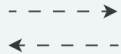
## Agent2Agent Protocol vs. Model Context Protocol

### Agent2Agent Protocol

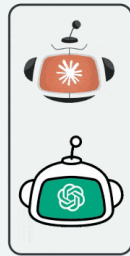
Source Agent



Agent2Agent  
Protocol



Target Agent



### Model Context Protocol

Source Agent



Model context  
Protocol (MCP)

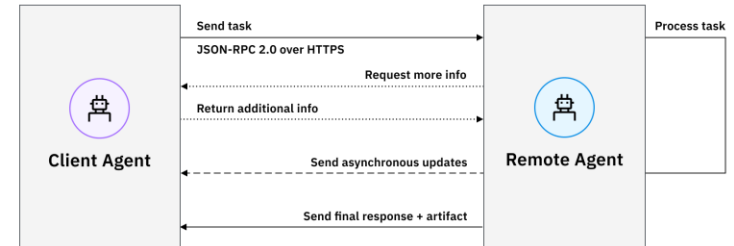


Tools



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### A2A Protocol



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# OOA – Object Oriented Agents

## NVIDIA-labs Object Oriented Agents

A Pythonic way to build AI agents.

[NVIDIA](#) [paper](#) [arXiv](#) [blog](#) [NVIDIA](#) [license](#) [Apache 2.0](#)

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NVIDIA-labs Object Oriented Agents (NOOA) is a model-agnostic Python framework designed to support reliable AI agent development. Many agent frameworks represent prompts, tools, callbacks, and workflows as separate abstractions. NOOA offers an alternative object-oriented interface that brings these concepts together in a Python class. NOOA lets developers express an agent's state, capabilities, prompts, and typed interfaces through a single Python class:

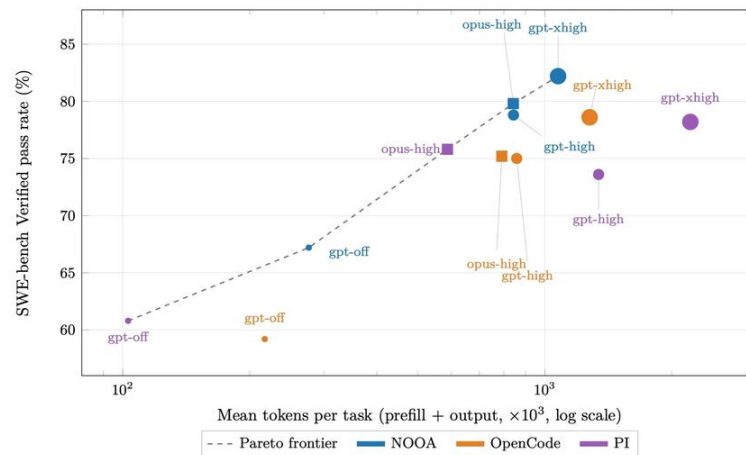
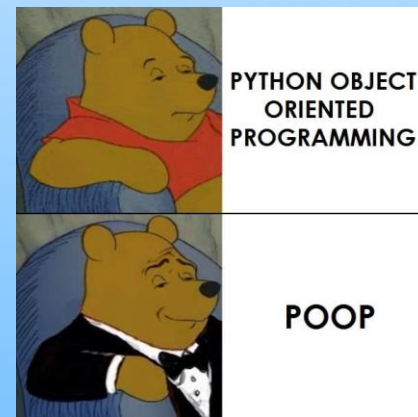


Figure 1. Score vs. total token cost per task on SWE-bench Verified. NOOA with GPT-5.5 reaches 82.2% at ~1.1M tokens per task. Data points represent published results at time of submission

# Wyzwania i ryzyka agentów



Meta's head of AI safety loses inbox as OpenClaw deletes emails by mistake: 'Ran like I was defusing bomb'



Johann Sathianathan  
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saas is dead

openclaw replaced all my subscriptions

went from \$480/month on tools to \$1,245/month on API costs & 15 hours a week fixing yaml files

adapt or be left behind

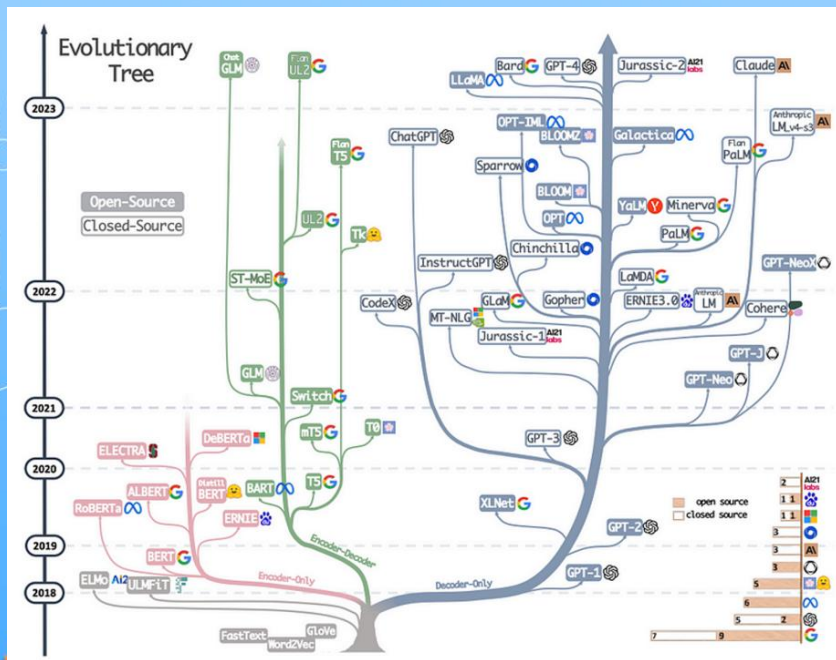


KN Solvro

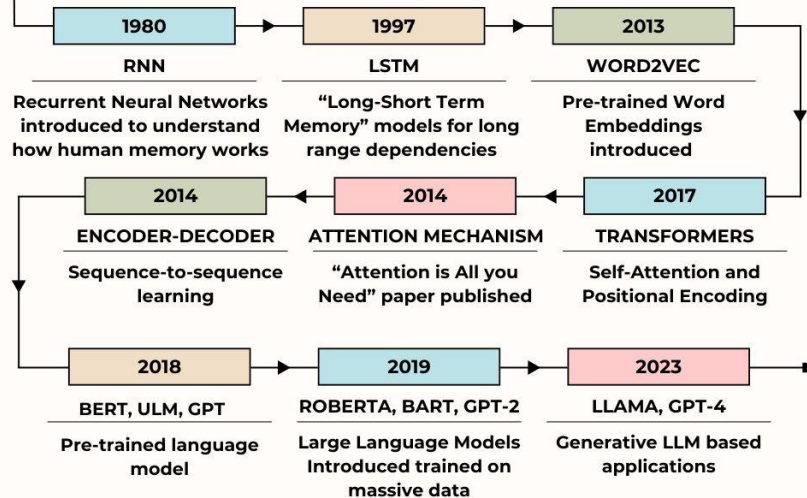




# Ewolucja NLP



## NLP EVOLUTION





bonus



# DZIĘKUJĘ ZA UWAGĘ !



## **Ekipa NLP :**

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KN Solvro